Science

Class 6 (Sindhi Medium)

Solved Exercises — STBB

12 lessons — 103 questions

Organization of Plant and Animal Cells

1. (a) Waste materials and useless things are removed from an animal's body through the nervous system.

1. (b) Hydra is a unicellular animal.

1. (c) The body-side lens (Objective) is used for focusing to see things magnified.

1. (d) Chromosomes are transferred from parents to offspring.

1. (e) In a system there is a group of tissues which together do the same kind of work.

2. (i) Which of the following absorbs the light of the sun?

(a) Cellulose

(b) Chlorophyll

(c) Cytoplasm

(d) Sap cell

2. (ii) Which of the following rows shows the wrong difference between plant and animal cells?

(a) Animal cell: It has no chloroplast. / Plant cell: It has chloroplast.

(b) Animal cell: It has many vacuoles. / Plant cell: It has one very large vacuole.

(c) Animal cell: It has a thin layer of cytoplasm. / Plant cell: Its space is filled with cytoplasm.

Question: 3. (i) Scientists use the microscope?

Answer: Scientists use the microscope because cells are so small that they cannot be seen with the naked eye; cells can only be seen with the help of a microscope.

Question: 3. (ii) Define the following and give examples: (a) Multicellular organisms (b) Organs

Answer: (a) Multicellular organisms: organisms whose body is made of more than one cell, for example humans, plants, eagles and Hydra. (b)

Organs: a group of different tissues doing the same kind of work which together do a particular job, for example the heart, liver, stomach, brain, leaf and flower.

Question: 3. (iii) Why are cells called the building blocks or units of life?

Answer: Cells are called the building blocks or units of life because all living things are made of cells and the cell is the basic unit of living things; all the processes of life take place in cells.

Question: 3. (iv) Make a chart showing the hierarchy from cell to organ and system which explains the digestive system of a human.

Answer: The hierarchy (in chart form): Cell → Tissue → Organ → System → Organism. Example (human digestive system): cells together form tissue, tissues together form an organ such as the stomach or intestines, different organs (mouth, tongue, stomach, liver, intestines) together form the digestive system, and all the systems together form the complete organism (human).

Sense Organs

1. (i) The ability to detect any stimulus of living things is called _____.

Answer: sense.

1. (ii) The cells which sense a particular kind of stimulus are called _____.

Answer: receptors.

1. (iii) A change felt in the environment of an organism is called _____.

Answer: a stimulus.

1. (iv) The human tongue can sense four kinds of tastes: _____, _____, _____ and _____.

Answer: sweet,, sour,, bitter, salty.

✓ 2. (i) Tell whether the following statement is true or false: Taste receptors are also present in the upper part of the mouth besides the tongue.

✗ 2. (ii) Tell whether the following statement is true or false: Light-sensing receptors are present in the cornea part of the eye.

✓ 2. (iii) Tell whether the following statement is true or false: The part of the brain in which smell is felt is called the olfactory bulb.

✗ 2. (iv) Tell whether the following statement is true or false: The eye is the sense of seeing (sight) and maintaining balance.

Question: 3. (a) Which of the stimuli given above is a pain stimulus? Which sense organ can sense it? (Stimuli: prick of a needle, hot gust of wind, your teacher's voice, fragrance of perfume, handshake)

Answer: The pain stimulus is the prick of a needle. The skin senses it, because pain-sensing receptors are present in the skin.

Question: 3. (b) Which of the stimuli given above is a temperature stimulus? And which sense will sense it? (Stimuli: prick of a needle, hot gust of wind, your teacher's voice, fragrance of perfume, handshake)

Answer: The temperature stimulus is the hot gust of wind. The skin will sense it, because heat- and cold-sensing receptors (thermo receptors) are

present in the skin.

Question: 3. (c) Which of the stimuli given above is a smell stimulus? And which sense or organ will sense it? (Stimuli: prick of a needle, hot gust of wind, your teacher's voice, fragrance of perfume, handshake)

Answer: The smell stimulus is the fragrance of perfume. The nose will sense it, because smell receptors are present in the nose.

Question: 3. (d) Which of the stimuli given above is a sound stimulus? And which sense or organ will sense it? (Stimuli: prick of a needle, hot gust of wind, your teacher's voice, fragrance of perfume, handshake)

Answer: The sound stimulus is your teacher's voice. The ear will sense it, because sound receptors are present in the ear.

Question: 3. (e) Which of the stimuli given above is a touch stimulus? And which sense or organ will sense it? (Stimuli: prick of a needle, hot gust of wind, your teacher's voice, fragrance of perfume, handshake)

Answer: The touch stimulus is the handshake. The skin will sense it, because touch/contact-sensing receptors are present in the skin.

Question: 4. (a) Make a flow chart and explain clearly through which stages a sound stimulus passes in reaching from the air to the brain.

Answer: Stages of the journey of sound (flow chart): sound waves in the air → outer ear or pinna (collects the sound) → ear canal → eardrum (vibrates with the sound) → malleus → incus → stapes (the three small bones pass the vibration onwards) → cochlea (the vibration causes movement in the fluid of the cochlea and the hair-like structures vibrate) → electrical signals through the cochlear nerve → brain (the brain understands the signals and produces the sensation of sound).

Question: 4. (b) Draw a neat, clean diagram of the human ear and label its different parts.

Answer: Parts of the diagram of the human ear (written description): Outer ear: pinna (the outer part of the ear which collects sound) and the ear canal (which carries sound inside). Middle ear: the eardrum and three small bones — malleus, incus and stapes. Inner ear: the cochlea (snail-like structure), the cochlear nerve and the semicircular canals for balance.

Question: 5. To which sense organ do the following structures belong? Cochlea, retina, lens, malleus, smelling or smell, tactile corpuscle, dermis

Answer: Cochlea — ear (part of the inner ear). Retina — eye. Lens — eye. Malleus — ear (one of the three small bones of the middle ear). Smelling or smell — nose (the work of sensing smell). Tactile corpuscle — skin (touch-sensing receptor). Dermis — skin (the middle layer of the skin in which the sensitive receptors are present).

Question: 6. Which of the following structures are present in which part of the body, and what work do they do? Pinna, lens, eardrum, smell receptors, taste buds

Answer: Pinna — present in the outer ear; it collects sound and sends it towards the ear canal. Lens — present in the eye; it focuses light and forms an image on the retina. Eardrum — present in the middle ear; it vibrates with sound waves and produces vibration. Smell receptors — present in the nasal cavity of the nose; they sense the chemical substances/smell present in the air. Taste buds — present on the tongue; they sense the tastes of the chemical substances present in food.

Question: 7. Tell in your own words how any sense organ of the body works.

Answer: Every sense organ of the body works like this: some stimulus from the environment (such as light, sound, smell, taste, touch, heat or pain) reaches the receptors present in the organ. The receptors sense that stimulus and make electrical signals. These signals reach the brain through the nerve, and the brain understands those signals and produces the sensation of that thing, such as the sensation of seeing, hearing, smelling, tasting or touching.

Question: 8. Explaining the working in the structure of the human eye, tell that it is also a kind of sense.

Answer: The eye is also a sense organ because it contains receptors which sense the stimulus of light. Structure and working of the eye: the cornea is the outer part of the eye which lets light enter the eye; the iris and the central hole (pupil) control the amount of light; the lens focuses the light and forms an image on the retina; the retina is the screen of the eye on which the light-sensitive receptors (photo receptors) are present, which sense the light and make electrical signals; the optic nerve carries these signals to the brain, and the brain understands them and produces the sensation of seeing. In this way the eye is a sense which senses the stimulus of light.

Photosynthesis and Respiration in Plants

1. (i) What work are the tiny pores or stomata in the leaves of a plant for?

✓ (a) For transpiration

(b) For absorption

(c) For transport

1. (ii) What is the green-coloured substance present in the leaf?

✓ (a) Chlorophyll

(b) Chloroplast

(c) Chromoplast

1. (iii) Which things are needed for the process of photosynthesis in plants?

(a) Water and oxygen

✓ (b) Water and carbon dioxide

(c) Water and sunlight

1. (iv) The process of photosynthesis lasts till daytime because it depends on:

(a) daytime activities

(b) daylight

✓ (c) sunlight.

1. (v) What do plants release at night?

✓ (a) Carbon dioxide

(b) Oxygen

(c) Water

Question: 2. (i) Which part of the plant is involved in the process of preparing food?

Answer: The leaf of the plant is involved in the process of preparing food. The chloroplasts present in the cells of the leaf (and the chlorophyll in them) prepare food (glucose) with the help of sunlight.

Question: 2. (ii) What are the functions of the pores (Stomata) in the leaf?

Answer: The function of the pores (stomata) in the leaf is the exchange of gases: through them carbon dioxide enters the leaf, and oxygen along with water vapour goes out.

Question: 2. (iii) What is the process of photosynthesis?

Answer: Photosynthesis is the chemical process in which plants, with the help of sunlight and chlorophyll, prepare their food (glucose) from water and carbon dioxide, and release oxygen. Chemical equation: $6CO_2 + 6H_2O$ — with the help of sunlight and chlorophyll → $C_6H_{12}O_6$ (glucose) + $6O_2$.

Question: 2. (iv) Tell why the process of photosynthesis is also very useful for living things.

Answer: The process of photosynthesis is very useful for living things because through this process plants prepare food and release oxygen. All living things depend on plants for food and oxygen; without plants, other living things could not get food, nor oxygen for breathing.

Question: 2. (v) Name the gases taking part in the process of photosynthesis and produced in the process.

Answer: The gas taking part in the process of photosynthesis is carbon dioxide (CO_2), and the gas produced in the process is oxygen (O_2).

Question: 2. (vi) What is the work of chlorophyll?

Answer: The work of chlorophyll is to absorb sunlight. Without chlorophyll the process of photosynthesis is not possible. It is chlorophyll which gives the leaf its green colour.

Question: 3. Match the following with their functions. (Broad structure of the leaf, tiny pores, root hairs, leaf cells, chlorophyll)

Answer: Broad structure of the leaf — it has tiny pores, that is, stomata. Tiny pores — taking in carbon dioxide and releasing oxygen. Root hairs — absorb mineral salts and water from the soil. Leaf cells — they contain solar panels. Chlorophyll — absorbs solar energy.

Question: Project: Make posters of the process of photosynthesis and the process of respiration and hang them in the class. If the process of photosynthesis stops in plants, what will happen?

Answer: Posters (written description): On the first poster show the process of photosynthesis — the leaf, sunlight, chlorophyll, and the equation: carbon dioxide + water — with the help of sunlight and chlorophyll → glucose + oxygen. On the second poster show the process of respiration — glucose + oxygen → carbon dioxide + water + energy. If the process of photosynthesis stops in plants, plants will not be able to prepare food and will not release oxygen; therefore the food of all living things and the oxygen for breathing will run out, and life will end.

Environment and Interaction With It

Question: Which of the following is a non-living (abiotic) part of the ecosystem?

Answer: (c) Mineral components — Mineral components are a non-living (abiotic) part, whereas bacteria, fungus and plants are living (biotic) parts.

Question: Which of the following is a living (biotic) part of the ecosystem?

Answer: (a) Bacteria present on your skin — Bacteria are living things, so they are a biotic part; minerals, water and temperature are abiotic parts.

Question: A tapeworm which lives inside the body of an organism and lives on that organism's food is:

Answer: (b) An example of a parasitic relationship. — In parasitism, one organism lives in or on the body of another organism and lives on its food, which harms the host organism.

Question: In any ecosystem, organisms can be divided into producers or consumers. Producers provide food for consumers. Name the organism from the following which makes producers and other consumers its food.

Answer: (b) Omnivore — eats all plants and animals — This organism eats both plants (producers) and other animals (consumers), so it makes both its food.

Question: Carbon is an essential part of the ecosystem. It keeps circulating through the whole ecosystem by being used and reused. Carbon is absolutely necessary to sustain life. Name the process in which carbon dioxide is used.

Answer: (b) The process of photosynthesis — In photosynthesis, plants use water and carbon dioxide to prepare their food with the help of sunlight.

Question: Photosynthesis is the chemical process that keeps happening in the leaf of a plant. But it needs a special kind of gas, which is also a non-living (abiotic) component of the environment. Tell me, which gas is this?

Answer: (a) Carbon dioxide — Plants absorb carbon dioxide from the air for photosynthesis. (The book prints "chlorophyll" in option (b), which is not a gas; chlorophyll is the name of the green substance present in leaves.)

Question: Chart of the mutual relationship between biotic and abiotic components — We need fuel for transportation. The waste materials released by every kind of transport affect the environment. Because of dependence on transport, you keep producing different waste materials in different amounts. With the help of the chart given below, find out and tell the amount of waste materials you produce in one week.

Answer: This is a practical activity; it has no single fixed answer. Complete it like this: for one week, write down daily which transport you used; against each day write the amount given in the book (walking — 0, cycling — 0, travelling by car — 200); then add the amounts of all seven days to get the total. For example, if the car was used three times in the week, the total waste gas would be 3 × 200 = 600. Conclusion: walking and cycling produce no waste gas, while travelling by car produces waste gas, so walking and cycling are better for the environment.

Atoms, Molecules, Mixtures and Compounds

Question: Which of the following particles carries a positive electric charge?

Answer: (a) Proton — A proton carries a positive electric charge, an electron carries a negative charge, and a neutron carries no charge.

Question: What is the chemical symbol for Manganese?

Answer: (a) Mn — The chemical symbol for manganese is Mn. (Mo is the symbol for molybdenum and Mg is the symbol for magnesium.)

Question: Write the names and symbols of the following elements in the given table.

Answer: Symbol | Name of element: Na — Sodium, Al — Aluminium, C — Carbon, Cl — Chlorine, O — Oxygen, H — Hydrogen, Ne — Neon, F — Fluorine, Ca — Calcium, N — Nitrogen.

Question: Tell two main differences between the following.

Answer: Metal and Non-Metal:

Question: Which methods (processes) would you use to separate the following mixtures (Mixtures) into their separate components? Describe these methods in detail.

Answer: (a) Mixture of sand and water — Filtration: Pass the mixture through filter paper; the sand will stay on the filter paper and the water will pass through below, so the two components will be separated.

Air

Question: Many gases are present in the air. Of all these gases, which gas is present in the largest amount?

Answer: (a) Nitrogen — Air contains 78% nitrogen, 21% oxygen and 1% other gases.

Question: What percentage of oxygen remains present in the air?

Answer: (c) 21% — The amount of oxygen in the air is 21%.

Question: Which gas stops the excess heat coming from the Sun?

Answer: (c) Ozone — The layer of ozone gas present in the atmosphere stops the intense heat and ultraviolet rays coming from the Sun from reaching us.

Question: What happens when oil burns?

Answer: (a) Carbon dioxide and energy are released. — In burning (combustion), the fuel reacts chemically with the oxygen in the air and produces carbon dioxide, water and energy.

Question: Tell two main functions of each of the following gases present in the air:

Answer: Oxygen:

Slurry and Suspension

Types of Energy

- Which of the following is an example of kinetic energy? (a) A moving motor car (b) A book being read (c) A compressed spring
- Which of the following is an example of potential energy? (a) A moving motor car (b) A compressed spring (c) A book being read
- A toy motor car moves, also makes a sound, and runs on a battery or cells. What kind of energy is there in the battery or cells? (a) Chemical energy (b) Kinetic energy (c) Thermal energy
- Imran and Ayesha are making a picture frame. Imran hammers a nail into the wood. Tell which kind of energy is NOT used in this work? (a) Electrical energy (b) Kinetic energy (c) Sound energy

Question: Briefly state the advantages of renewable (reusable) sources of energy.

Answer: (1) These sources never run out — the sun, wind and water are always present. (2) They do not pollute the environment and are harmless. (3) A large amount of energy is obtained from them. (4) Using them keeps the earth and the environment safe and reduces dependence on fossil fuels.

Question: Write any three differences between kinetic and potential energy in the table below. | Kinetic energy | Potential energy |

Answer: 1. Kinetic energy is obtained from motion. | Potential energy is energy collected (stored) in a body. 2. It depends on the mass of the body and the speed of its motion. | It depends on the mass of the body and its height (position) above the ground. 3. In it, the body is in motion. | In it, the body is completely at rest.

Force and Machines

- Which of the following simple machines is a wheel and axle? (a) A knife (b) The steering of a motor car (c) A swing (d) A bottle opener
- Gears do all of the following EXCEPT which one? (a) Changing speed (b) Changing direction (c) Changing the fulcrum (d) Changing the amount of force
- What can a movable pulley do? (a) It increases the applied force (b) It decreases the applied force (c) It can change the direction of the force (d) It can change the speed of the force
- In which of the following is a movable pulley used? (a) In flags (b) On wells (c) In cranes (d) In bicycles
- Which simple machine is shown in the picture of the red wagon? (a) A screw (b) A pulley (c) An inclined plane (d) Wheel and axle

Question: Write the names of the machines that work in a bicycle.

Answer: 1. Gears 2. Wheel and axle (the wheels) 3. Pulley system (the chain) 4. Lever — the pedals 5. Lever — the brake/handlebar.

Question: Wheel and axle is a type of lever. Explain.

Answer: Wheel and axle is a special type of lever. The wheel acts like the arm of the lever, and the centre of the wheel works as the fulcrum. Turning the wheel also turns the axle, and in this way a weight or load can be moved or lifted easily.

Question: State two differences between a fixed and a movable pulley.

Answer: 1. A fixed pulley stays fixed at one place and does not move with the load; a movable pulley moves up/down along with the load. 2. A fixed pulley does not change the amount of force, it only changes the direction of the force (applying force downward lifts the load upward); a movable pulley increases the applied force, so a large load can be lifted with less force than the weight of the load.

Question: Look at the diagram of gears given below. Gear (A) turns in the clockwise direction. State the direction of turning of the other gears.

Answer: Meshed gears always turn in opposite directions: gear (A)

clockwise → gear (D) anticlockwise → gear (C) clockwise → gear (B) anticlockwise; and from gear (D), gear (E) clockwise → gear (F) anticlockwise.

Question: Explain the working method of a bicycle.

Answer: The rider pushes the pedals; the pedals drive the chain; the chain turns the gear of the back wheel; the gear turns the wheel and the bicycle moves forward. The handlebar (lever) changes the direction, and the brake slows or stops the bicycle. Changing gears controls the force and the speed.

Question: Clearly state the work of the following simple machines: 1. Pulley 2. Gear 3. Wheel and axle

Answer: 1. Pulley — lifting a load upward, changing the direction of force, and increasing force; examples: drawing water from a well, raising a flag, lifting heavy goods in a crane. 2. Gear — changing the amount of force, speed and direction; examples: bicycle, motor car, clock. 3. Wheel and axle — turning a weight easily and covering a long distance; examples: wheels of vehicles, bicycle, fan, water pump.

Question: Find at least one example of each of the following things used in your home: (i) an example of each pulley (ii) wheel and axle (iii) gears. Make pictures or sketches of these things and write the names of their parts. Also tell in which jobs these machines are used and why they are simple machines.

Answer: (i) Pulley — the pulley of a well, a curtain pulley or a flag pulley; use: drawing water/raising a curtain, changing the direction of force. (ii) Wheel and axle — a door handle, a tap, a fan or a bicycle wheel; use: making work easy by turning. (iii) Gears — the gears of a clock or of a bicycle; use: controlling speed and force. They are simple machines because they have few parts and they make work easier — they increase force, change the direction of force, or change speed. (Pictures/sketches: the student draws pictures of the home items and labels their parts.)

Properties of Light

1- Fill in the blanks: (a) Light travelling in a straight line, when it strikes another medium, _____ . (b) The image formed by a plane mirror is _____ the object. (c) _____ type of mirrors are used for shaving and make-up. (d) In submarines, _____ is used to see things on the surface of the sea. (e) In commercial and business centres, _____ mirrors are used as safety mirrors.

Answer: returns back., equal in size to, Concave, a periscope, convex

Question: 2- Divide these into two groups - "light-giving sources" and "non-light-giving sources": stars, Moon, Sun, mirror, torch, diamond (Diamond), table lamp, chalk, burner mate light, wach ya knoit, table, television (switched on or running).

Answer: Light-giving sources: stars, Sun, torch, table lamp, burner mate light, wach ya knoit, television (switched on or running). Non-light-giving sources: Moon, mirror, diamond, chalk, table. (Explanation: light-giving sources produce their own light; non-light-giving sources only reflect the light of others — for example, the Moon reflects the Sun's light and a mirror reflects light.)

Question: 3- What are the absorption of light and the reflection of light?

Answer: The absorption of light is the process in which, when light falls on an object, part of it is absorbed into the object, and the absorbed light is converted into heat, so that object becomes warm. The reflection of light is the process in which, when light strikes a shiny or smooth surface, it turns back from its original path — that is, it returns from the surface.

Question: 4- Which of the surfaces: (a) give regular reflection of light (b) give irregular reflection of light?

Answer: (a) Smooth and shiny surfaces, such as a mirror, still water and polished metal, give regular reflection of light. (b) Rough and uneven surfaces, such as paper, cloth, a wall and wood, give irregular reflection of light. (Note: the book gives no list of surfaces with this question, so the answer is given from the general rule in the chapter.)

Question: 5- State the uses of convex and concave mirrors.

Answer: Concave mirror: it is used in the headlights of vehicles, in the dentist's mirror, in telescopes, and in shaving and make-up mirrors, because it gives a large, upright and virtual image of a nearby object. Convex mirror: it is used in motor cars to see the rear view, and as a safety mirror in shops, commercial centres and on roads, because it shows a wide view.

Question: 6- State the properties and important points of the image formed by a plane mirror.

Answer: The properties of the image formed by a plane mirror are: (1) The image is virtual, meaning it appears behind the mirror but is not actually present there. (2) The image is upright. (3) The image is the same size as the object. (4) The image is as far behind the mirror as the object is in front of it. (5) In the image, the right and left sides appear reversed, which is why the word AMBULANCE is written reversed on an ambulance, so that a driver ahead sees it straight in his mirror.

Question: 7- When light reflects from a plane mirror, what is the important point about the angle of the ray coming towards the mirror and the angle of the ray returning from the mirror?

Answer: The law of reflection of light is that the angle of the ray coming towards the mirror (angle of incidence i) is equal to the angle of the ray returning from the mirror (angle of reflection r), that is, i = r.

Question: 8- Why does a diamond appear sparkling from many directions and in many colours?

Answer: The text of this chapter does not explain why a diamond sparkles from many directions and in many colours, so this question cannot be answered from the chapter. The chapter only states that light reflects from shiny surfaces and can reflect more than once (multiple reflections).

Question: 9- Why do you not use a concave mirror to see the rear view in a motor car?

Answer: A concave mirror does not work for seeing the rear view, because when the object is far from it, its image is real, inverted and much smaller than the actual object, and its field of view is also narrow. In contrast, a convex mirror shows the view over a wide distance, which is why a convex mirror is used in motor cars to see the rear view.

Question: 10- If a driver uses a plane mirror and a convex rear-view mirror in his vehicle, how will the images formed in the two mirrors differ from each other?

Answer: The image formed in the plane mirror is the same size as the object, upright and virtual. The image formed in the convex mirror differs from it in that the rear view appears wider in it — that is, a larger area is visible in the same mirror — and so the objects behind appear smaller.

Information About Sound

Question: 1- Put a (✓) mark on the correct answer among the answers to these questions. (i) Through which of these things can sound pass fastest? (a) Through sea water (b) Through air (c) Through gold (d) Through space (ii) When sound passes through air, the particles of air: (a) vibrate together with the moving waves of sound. (b) do not vibrate in any fixed direction. (c) vibrate in a direction perpendicular to the waves. (d) keep vibrating in straight lines. (iii) Sound waves first enter into what? (a) Into the eardrum (b) Into the long canal of the ear (c) Into the cochlea (d) Into the stirrup (Stirrup) (iv) What is the speed of sound in air? (a) 345 metres per second (b) 333 metres per second (c) 332 metres per second (d) 354 metres per second (v) When sound waves reach the eardrum: (a) the sound is felt by the brain. (b) the hair cells present in the cochlea (Cochlea) can understand the vibration coming from the eardrum (c) the hearing nerve feels the vibration and sends it to the brain (d) they send the vibration of the small bones behind the eardrum towards the fluid of the inner ear

Answer: (i) ✓ (c) Through gold (according to the chapter's table, the speed of sound in gold is 3240 metres per second, which is more than sea water 1530 and air 332, and sound cannot pass through space at all.) (ii) ✓ (a) They vibrate together with the moving waves of sound. (iii) ✓ (b) Into the long canal of the ear. (iv) ✓ (c) 332 metres per second. (v) ✓ (d) They send the vibration of the small bones behind the eardrum towards the fluid of the inner ear.

Question: 2- Complete the path of sound waves from the ear canal to the brain in the diagram chart given below. Ear canal ← [ ] ← [ ] ← [ ] ← [ ] ← Brain

Answer: Ear canal ← eardrum (the drum of the ear) ← three small bones ← cochlea ← hearing nerve ← brain. (Explanation: sound waves first pass through the ear canal and vibrate the eardrum; the vibration of the eardrum reaches the fluid of the cochlea through the three small bones; the fine hairs of the cochlea feel the vibration and send it to the brain through the hearing nerve, and then we hear the sound.)

Question: 3- In which parts of the things given below is vibration produced? (Pictures: drum, speaker, guitar)

Answer: Drum: vibration is produced in its skin membrane (cover). Speaker: vibration is produced in its diaphragm (cone). Guitar: vibration is produced in its strings.

Question: 4- (a) Tell what sound is and how it is produced. (b) With the help of a diagram, explain clearly how the compressed and rarefied waves of sound are produced near the source of sound.

Answer: (a) Sound is also a form of energy. Sound is produced when a body vibrates — that is, moves forwards and backwards. (b) When the

source of sound (such as the diaphragm of a speaker) moves forwards, it pushes the air particles in front of it and gathers them together, which produces a compression where the air particles gather. When the source moves backwards, the air particles move away from each other, which produces a rarefaction. With the continuous forward and backward vibration of the source, a series of compressions and rarefactions keeps being produced one after another, and these are the longitudinal waves of sound which travel through the air and reach our ears.

5- Write a one-word answer for these: (a) The very fast forward and backward movement of a thing _____. (b) The place in a sound wave where the particles of air are very close to each other _____. (c) The place in a sound wave where the particles of air are very far from each other _____. (d) What is the name of the last of the three small bones inside the ear?

Answer: Vibration., Compression., Rarefaction.

Question: 6- Additional activity: Can sound make the spot of light dance? (i) Take a personal round box and open both its ends to make a cylinder. (ii) Cut a balloon and fix it on one side of this cylinder, and hold it in place with a rubber band. (iii) Cut a small piece of mirror, put a drop of glue on it, and stick it on top of the balloon layer. (iv) Fix the box on a table or a stand. (v) Throw the ray of light from a slit (Slit) onto the mirror. (vi) Turn the light towards the wall and a small circle of light will be seen there. (vii) Speak loudly from the other open side of the box and you will see the moving (dancing) spot of light on the wall. (viii) Discuss with your friends and explain clearly why the circle of light can move or dance?

Answer: Yes, sound can make the spot of light dance. When you speak loudly from the open side of the box, the sound waves vibrate the balloon layer, and along with it the small mirror stuck on it also starts vibrating. Because of the vibration of the mirror, the ray of light reflecting from it also starts moving, so the circle (spot) of light on the wall appears to dance. This activity proves that sound is produced by vibration, and that sound waves can make other things vibrate too.

Question: Project: Have you ever thought about how a guitar (Guitar) plays? Through this scientific experiment we will find out how a melodious sound is produced from a guitar. (Things needed: a shoe box for keeping water, elastic strips or rubber bands, drawing pins (Pins), a wooden strip, glue (Glue) or gum, and scissors. Method: 1- Cut an empty shoe box. Make a round circle of 8 centimetres diameter on one side of it and make a round hole around it. This is called the sound hole (Sound Hole), and it is generally in the middle of the sound box (Sound Box) of the guitar. 2- Find rubber strips of different thicknesses, arranged from thick to thin, and cut them. Fix them on the sides of the box with the help of pins, as shown. 3- Put the wooden strip under the rubber strips with glue, as shown. 4- Now you are fully ready to use the guitar.)

Answer: A guitar plays when its strings (the rubber strips) are plucked: they vibrate, and sound is produced by the vibration. In this project, the shoe box does the work of the guitar's sound box (Sound Box), and the round hole made in it is the sound hole (Sound Hole). Rubber strips of different thicknesses are stretched and fixed on the box with pins, and a wooden strip is glued underneath. When the rubber strips are plucked,

they vibrate, and their vibration produces a melodious sound.

Space and Satellites

1- Complete these sentences with the help of the given words: [Meteors, Mars, Asteroids, Natural, Global, Jupiter, Sun, Geostationary, Photograph, Navigation.] (i) The Moon is the _____ satellite of the Earth. (ii) In the solar system, many _____ are present in the orbits and _____. (iii) The revolving time of 24 hours is that of _____. (iv) A shooting star is called _____. (v) _____ type of satellites are used to find the location of ships, aeroplanes and motor vehicles. (vi) A comet is seen when it passes near the _____. (vii) In the word GPS, G is used for _____. (viii) With the help of satellites, _____ of the Earth are taken for observing the Earth. (ix) The first space artificial satellite was sent into space by _____.

Answer: natural, asteroids, Jupiter., the geostationary satellite., a meteor., Navigation, Sun., Global., photographs, Russia. (Note: the order of the printed sentence in (ii) is misprinted in the book; the chapter's meaning is that asteroids revolve between Mars and Jupiter. The answer 'Russia' in (ix) is neither among the given words nor stated in the text of this chapter.)

Question: 2- Put a mark of 'T' if the following sentences are correct and 'F' if they are wrong: (i) Navigation satellites are used to broadcast T.V. signals. F/T (ii) Artificial satellites are man-made. F/T (iii) Only the Moon is the natural satellite of the solar system. F/T (iv) Meteors are small asteroids made of stones. F/T (v) Natural satellites are not celestial bodies. F/T (vi) Small asteroids (Asteroids) are called minor planets (Minor-Planets). F/T (vii) The satellite named Landsat was sent by Pakistan. F/T (viii) Comets appear in the sky like a streak of light. F/T

Answer: (i) ✗ F — Broadcasting T.V. signals is the work of communication satellites, not navigation satellites. (ii) ✓ T (iii) ✗ F — 173 natural satellites have been discovered in space so far, not only the Moon. (iv) ✓ T (v) ✗ F — Natural satellites are celestial bodies. (vi) ✓ T (vii) ✗ F — Landsat is a satellite of the USA. (viii) ✓ T

Question: 3- Give short answers to these questions: (i) Explain the meaning and concept of a satellite, and also state its types. (ii) State the names of the natural satellites of the Sun. (iii) What is a comet, and how is it different from a small asteroid? (iv) With the help of an artificial satellite, how do we find the location of a thing on the Earth? (v) Into how many types can artificial satellites be divided according to their work? State only their names.

Answer: (i) A satellite is that celestial body which revolves around another bigger celestial body — that is, a planet or a star. There are two types of satellites: (1) Natural satellites — which exist naturally, such as the Moon; (2) Artificial satellites — which man makes and sends into space. (ii) The natural satellites (planets) of the Sun are: Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus, Neptune and Pluto (a minor planet). (iii) A comet is actually a ball of ice made of mixed particles of ice, dust and gases; when it comes near the Sun, its coma and tail become visible, and it revolves

around the Sun in an elliptical orbit. In contrast, a small asteroid is an irregularly shaped body made of rock and metal, which has no tail, and asteroids mostly revolve between Mars and Jupiter. (iv) Signals are sent from a group of artificial satellites (the GPS system); a receiver on the Earth receives those signals, calculates, and tells the exact location of any thing or person. (v) There are five types of artificial satellites: (1) Weather satellites, (2) Communication satellites, (3) Earth observation satellites, (4) Navigation and marine travel satellites, (5) Scientific satellites.

Question: 4- Recognise the marked objects (Objects) in the picture given below well, and write them in the correct answer box, choosing from the words given below. (Meteor, Comet, Asteroid belt) (The picture has 1, 2, 3, 4 marked on it.)

Answer: 1. Asteroid belt (a cluster of rocks in space). 2. Meteor (a rock floating in space — a meteoroid). 3. Meteor (a rock burning in the Earth's atmosphere — a shooting star). 4. Meteor (a rock that has reached the Earth — a meteorite). (Note: of the three words given in the book, 'comet' is nowhere among the marked objects — the comet drawn at the side of the picture is only a hint. In the chapter's own words, the rock in space, the burning rock and the rock that has reached the Earth are all called 'meteors'.)

Question: 5- Circle the correct answer among the answers to these questions: (i) Which of these is not a natural satellite (Satellites)? (a) Moon (b) Sputnik (Sputnik-1) (c) Saturn (d) Comet (Comet) (ii) What are Earth observation satellites used for? (a) For broadcasting signals (b) For preparing maps (c) For communication (d) For space research (iii) What is the Global Positioning System (GPS) used for? (a) For sending artificial satellites into space (b) For forecasting the weather (c) For finding the location of things on the Earth (d) For obtaining information about the Sun and other planets

Answer: (i) ✓ (b) Sputnik (Sputnik-1) — it is a man-made artificial satellite. (ii) ✓ (b) For preparing maps. (iii) ✓ (c) For finding the location of things on the Earth.

Question: 6- To show the properties of small asteroids (Asteroids), comets (Comets) and meteors (Meteors), put a (✓) mark in the correct boxes. (Properties: 1. It is seen in the sky like a streak of light. 2. It is a ball of ice like a ball mixed with dust (Dust). 3. It is seen in the sky even without the help of any instrument. 4. It is made of a rocky body. 5. It looks like a hairy star. 6. It keeps revolving around the Sun. 7. Halley's comet is an example of it. 8. They are leftover, useless, tiny particles of planets. 9. It is generally called a shooting star. 10. It is mostly completely destroyed on entering the Earth's atmosphere. 11. On coming near the Sun, its tail is seen. 12. They are mostly present in the orbits of Mars and Jupiter. 13. Sometimes it is seen in the sky in the shape of a bow. 14. It takes many years to complete one revolution around the Sun.)

Answer: 1. Like a streak of light — Comets ✓, Meteors ✓. 2. Ball of ice — Comets ✓. 3. Seen without an instrument — Comets ✓, Meteors ✓. 4. Rocky body — Asteroids ✓, Meteors ✓. 5. Like a hairy star — Comets ✓. 6. Revolves around the Sun — Asteroids ✓, Comets ✓. 7. Halley's comet — Comets ✓. 8. Leftover tiny particles — Meteors ✓. 9. Shooting star — Meteors ✓. 10. Destroyed in the atmosphere — Meteors ✓. 11. Tail seen

near the Sun — Comets ✓. 12. In the orbits of Mars and Jupiter — Asteroids ✓. 13. In the shape of a bow — Comets ✓. 14. Many years for one revolution — Comets ✓. (Note: the revolving of meteors around the Sun in property 6 is not stated in the chapter, so it has not been marked. Property 13 is unclear in the book; the curved tail of a comet looks like a bow.)

Question: 7- Look at these pictures well and put labels of their names on them: (Pictures: astronaut, artificial satellite, 24-hour orbit, rocket)

Answer: Picture of the astronaut: Astronaut. Picture of the satellite: Artificial satellite (Satellite). Picture of the 24-hour orbit: Geostationary satellite. Picture of the rocket: Rocket (Rocket).

Question: Project: Prepare a model of your own artificial satellite (Satellite). (Things needed: white card sheet, wooden board or table, glue (Glue) or gum, scissors and a paper-cutting knife. Steps of the project: 1- Carefully separate the cuttings (1, 2 and 3) given in the pictures. 2- Cut picture number 3 and, with the help of a paper cutter (knife), make small cuts (Cuts) at the places A, B, C and D. 3- Now join those cuts or marks in picture 3 with the different parts of the satellite given in pictures 1 and 2. 4- Fold the sheet along the dashed lines (Dashed Lines) in picture number 3, so that you get a well-shaped structure. 5- In the end, join the cuts of the body of the rocket with glue (Glue). Now your satellite is fully ready for flight. (1) Solar panel: it is used to provide energy or fuel (Fuel) to this satellite. (2) Satellite dish antenna: it is used to give information to the Earth and to obtain information.)

Answer: The model of an artificial satellite is prepared like this: first the parts (1, 2 and 3) given in the pictures are carefully cut from the card sheet, then small cuts are made at the places A, B, C and D in picture number 3. After that, those cuts are joined with the different parts of the satellite in picture 3, and the sheet is folded along the dashed lines so that the correct shape of the satellite is formed. In the end, the cuts of the body of the rocket are joined with glue, and the model is ready. In the model, (1) the solar panels provide energy (fuel) to the satellite, and (2) the satellite dish antenna does the work of giving information to the Earth and obtaining information from the Earth.